

**Department: Computer Engineering**

**PRACTICAL PLAN**

| <b>A. Y. 2025-26</b>             |                                              |                         |                     |
|----------------------------------|----------------------------------------------|-------------------------|---------------------|
| <b>Subject Name:</b>             | <b>Database Management System Laboratory</b> | <b>Course Pattern:</b>  | <b>2024</b>         |
| <b>Subject Short &amp; Code:</b> | <b>DBMSL - PCC-254-COM</b>                   | <b>Division(s):</b>     | <b>A/B</b>          |
| <b>Class &amp; Semester:</b>     | <b>SE, IV</b>                                | <b>Pract.Hrs./week:</b> | <b>02</b>           |
| <b>Faculty Name:</b>             | <b>Mr. R. S. Jagale / Ms. J. S. Pawar</b>    | <b>w.e.f.</b>           | <b>01/ 01 /2026</b> |

**Course Outcomes:**

| <b>COs</b> | <b>CO Statement</b>                                                                                                                                                |
|------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| CO1        | To understand and apply the concepts of database design by formulating case studies, creating E-R diagrams, and mapping them to the relational model.              |
| CO2        | To develop and execute SQL queries for creating, modifying, and managing database structures using DDL, DML, DCL, and TCL commands                                 |
| CO3        | To implement advanced SQL operations, including aggregate functions, joins, subqueries, views, stored procedures, and triggers, for efficient database management. |
| CO4        | To explore NoSQL databases by designing and implementing CRUD operations in MongoDB, understanding document-based storage and retrieval.                           |

| <b>Asst. No.</b> | <b>Contents</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | <b>COs</b> |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|
| 1.               | <b>Case Study and ER Diagram</b><br>Develop a case study and design its Entity-Relationship (ER) Diagram. Convert the ER model into a relational model.                                                                                                                                                                                                                                                                                                                                                                     | CO1        |
| 2.               | <b>Write and execute SQL Data Definition Language (DDL)</b> commands such as CREATE, ALTER, DROP, RENAME, and TRUNCATE to define and modify tables. Insert data into the tables and apply appropriate integrity constraints such as NOT NULL, UNIQUE, PRIMARY KEY, FOREIGN KEY, and CHECK.<br>(The application may vary as per the subject teacher's requirement.)                                                                                                                                                          | CO2        |
| 3.               | <b>SQL Queries for Data Manipulation, Access Control, and Transactions</b><br>Design and run SQL queries to demonstrate the following: a) Data Manipulation (DML): Use SQL statements to INSERT, UPDATE, and DELETE records. Apply arithmetic, logical, set operators, pattern matching, and string functions. b) Access Control (DCL): Use GRANT, REVOKE, and ROLE commands to manage user access. c) Transaction Control (TCL): Apply START TRANSACTION, COMMIT, ROLLBACK, and SAVEPOINT commands to manage transactions. | CO2        |

|     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                          |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------|
| 4.  | <b>Aggregate Functions and Grouping</b><br>Use aggregate functions along with GROUP BY and HAVING clauses to retrieve summarized data from the database                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | CO3                      |
| 5.  | <b>JOIN Operations and Views</b><br>Perform various types of JOIN operations to extract meaningful relationships between tables. Create and manage different database views.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | CO3                      |
| 6.  | <b>Subqueries</b><br>Write and execute subqueries to retrieve data from one table based on results from another.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | CO3                      |
| 7.  | <b>Stored Procedures or Function with Cursors</b><br>Create and execute stored procedures / functions using cursors.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | CO3                      |
| 8.  | <b>Database Triggers</b><br>Implement and test triggers to maintain data integrity in the database.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | CO3                      |
| 9.  | <b>CRUD Operations using MongoDB</b><br>Design and implement basic Create, Read, Update, and Delete (CRUD) operations using MongoDB. Use the save method and logical operators where necessary                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | CO4                      |
| 10. | <b>Aggregation and Indexing in MongoDB</b><br>Design and execute MongoDB queries using aggregation and indexing techniques with suitable examples.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | CO4                      |
| 11. | Using the database concepts covered in above assignments, develop an application with following details:<br>1. Follow the Software Development Life cycle and other concepts learnt in Software Engineering Course throughout the implementation.<br>2. Develop application considering:<br>• Front End: Java/Perl/PHP/Python/Ruby/.net/any other language<br>• Backend : MongoDB/ MySQL/Oracle<br>3. Test and validate applications using Manual/Automation testing.<br>4. Student should develop application in group of 2-3 students and submit the Project Report which will consist of documentation related to different phases of Software Development Life Cycle:<br>• Title of the Project, Abstract, Introduction<br>• Software Requirement Specification<br>• Conceptual Design using ER features, Relational Model in appropriate Normalize form<br>• Graphical User Interface, Source Code<br>• Testing document<br>• Conclusion | CO1<br>CO2<br>CO3<br>CO4 |

Date: 01/01/2026

Mr. R.S.Jagale / Ms. J. S. Pawar  
Prepared by  
**Subject In-charge**

Dr. D. V. Patil  
Approved by  
**HOD**